Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-04-12 10:43:09

Situation Summary

Stage Iv: 18 of 88
Biomarker Count: 88 entries
Top Targets: RESEARCH (31) · GENOMIC PROFILING (28) · LIQUID BIOPSY (23)
Breakthrough: 35 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

MEDIUM · Stage IV · GENOMIC PROFILING · VERIFIED · GENOMIC PROFILING · PUBLISHED MAR 28, 2026 · 9

The Evolving Role for Repeat Molecular Testing in Metastatic Colorectal Cancer

This study evaluates the evolving role for repeat molecular testing in metastatic colorectal cancer.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM · Stage IV · GENOMIC PROFILING · VERIFIED · GENOMIC PROFILING · PUBLISHED APR 01, 2026 · 8

Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

This study evaluates the clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

HIGH · All Stages · LIQUID BIOPSY · VERIFIED · LIQUID BIOPSY · PUBLISHED APR 03, 2026 · 7

Liquid biopsies using circulating tumor DNA for surveillance of gastrointestinal cancers in Hispanics: first real-world data report

This study evaluates liquid biopsies using circulating tumor DNA for surveillance of gastrointestinal cancers in Hispanics.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

HIGH · All Stages · LIQUID BIOPSY · VERIFIED · LIQUID BIOPSY · PUBLISHED MAR 30, 2026 · 6

Utility of Urinary Extracellular Vesicles for Colorectal Cancer: Comparison of DNA Quality and MRD Detection

This study evaluates the utility of urinary extracellular vesicles for colorectal cancer through a comparison of DNA quality and MRD detection.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

HIGH All Stages LIQUID BIOPSY VERIFIED LIQUID BIOPSY PUBLISHED MAR 14, 2026 6

Multistep ctDNA Monitoring of Minimal Residual Disease in Colorectal Cancer Liver Metastases: From Tissue NGS to Highly Sensitive Digital PCR Platforms

This study evaluates multistep ctDNA monitoring of minimal residual disease in colorectal cancer liver metastases, from tissue NGS to highly sensitive digital PCR platforms.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

HIGH Early Detection LIQUID BIOPSY VERIFIED LIQUID BIOPSY PUBLISHED MAR 25, 2026 4

Blood-based circulating tumour DNA (ctDNA) tests for colorectal cancer screening: Systematic review and meta-analysis of diagnostic accuracy

This study evaluates blood-based circulating tumour DNA (ctDNA) tests for colorectal cancer screening through a systematic review and meta-analysis of diagnostic accuracy.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

HIGH Early Detection LIQUID BIOPSY VERIFIED LIQUID BIOPSY PUBLISHED APR 01, 2026 3

Circulating Tumor Cells as the Liquid Biopsy Foray into Noninvasive Colorectal Cancer Screening

This study evaluates circulating tumor cells as the liquid biopsy foray into noninvasive colorectal cancer screening.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages GENOMIC PROFILING VERIFIED GENOMIC PROFILING PUBLISHED MAR 14, 2026 3

Dual testing with immunohistochemistry and PCR for enhanced accuracy in dMMR/MSI detection in Greek colorectal cancer population: the impact of retesting

This study evaluates dual testing with immunohistochemistry and PCR for enhanced accuracy in dMMR/MSI detection in the Greek colorectal cancer population and the impact of retesting.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages RESEARCH VERIFIED RESEARCH PUBLISHED MAR 31, 2026 2

BPI26-006: Too Late to Test: Real-World Impact of MSI/MMR Testing Delays on Survival and Treatment in Colorectal Cancer

This study evaluates the real-world impact of MSI/MMR testing delays on survival and treatment in colorectal cancer.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages RESEARCH VERIFIED RESEARCH PUBLISHED MAR 18, 2026 2

Molecular Profiling of Germline Variants in the DNA Mismatch Repair Genes in Chinese Colorectal Cancer Patients

This study evaluates molecular profiling of germline variants in the DNA mismatch repair genes in Chinese colorectal cancer patients.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages GENOMIC PROFILING VERIFIED GENOMIC PROFILING PUBLISHED MAR 17, 2026 2

A paradigm shift in genetic predisposition to colorectal cancer: the impact of germline multigene panel testing on diagnosis and management

This study evaluates the impact of germline multigene panel testing on diagnosis and management in the context of genetic predisposition to colorectal cancer.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages RESEARCH VERIFIED RESEARCH PUBLISHED MAR 16, 2026 2

Genetic Alterations Involved in Immune Escape Mechanisms of Circulating Tumour Cells in Colorectal Carcinogenesis

This study evaluates genetic alterations involved in immune escape mechanisms of circulating tumour cells in colorectal carcinogenesis.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection EARLY DETECTION VERIFIED EARLY DETECTION PUBLISHED APR 01, 2026 1

From mono- to multi-cellular in vitro models: reconstructing colorectal cancer complexity for translational and personalized applications

This study evaluates the transition from mono- to multi-cellular in vitro models for reconstructing colorectal cancer complexity for translational and personalized applications.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection GENOMIC PROFILING VERIFIED GENOMIC PROFILING PUBLISHED MAR 27, 2026 1

Biomarkers in Colorectal Cancer: Clinically Relevant Diagnostic and Prognostic Molecular Features, and the Future of Precision Medicine

This study evaluates biomarkers in colorectal cancer, focusing on clinically relevant diagnostic and prognostic molecular features, and the future of precision medicine.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages RESEARCH VERIFIED RESEARCH PUBLISHED MAR 26, 2026 1

A PMS2-deficient pediatric high-grade glioma with PI3K-pathway mutations and adjacent developmental venous anomaly suggestive of CMMRD

This study evaluates a PMS2-deficient pediatric high-grade glioma with PI3K-pathway mutations and adjacent developmental venous anomaly suggestive of CMMRD.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages RESEARCH VERIFIED RESEARCH PUBLISHED MAR 24, 2026 1

dMMR prediction from colorectal cancer histopathology: Leveraging non-tumor and low-magnification regions

This study evaluates dMMR prediction from colorectal cancer histopathology by leveraging non-tumor and low-magnification regions.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages RESEARCH VERIFIED RESEARCH PUBLISHED MAR 23, 2026 1

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

This study evaluates how circulating butyrate attenuates cetuximab efficacy in colorectal cancer through EGFR and AMPK-Wip1 signaling.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages GENOMIC PROFILING VERIFIED GENOMIC PROFILING PUBLISHED MAR 13, 2026

TP53 Loss Fuels mTORC1 Activation and Autophagy Suppression to Drive Immune-Cold Colorectal Cancer ⚡ 1

This study evaluates how TP53 loss fuels mTORC1 activation and autophagy suppression to drive immune-cold colorectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM | All Stages | RESEARCH | VERIFIED | RESEARCH | PUBLISHED APR 09, 2026 ⚡ 0

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

This study evaluates the integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible multi-outcome prediction in a multi-center study of colorectal cancer.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM | All Stages | RESEARCH | VERIFIED | RESEARCH | PUBLISHED APR 08, 2026 ⚡ 0

An algorithm-enhanced stool DNA system improves the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients

This study evaluates an algorithm-enhanced stool DNA system that improves the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM | Stage III | RESEARCH | VERIFIED | RESEARCH | PUBLISHED MAR 30, 2026 ⚡ 0

Identifying gene expression signatures for risk stratification of postoperative adjuvant chemotherapy in colorectal cancer

This study evaluates gene expression signatures for risk stratification of postoperative adjuvant chemotherapy in colorectal cancer.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM | All Stages | RESEARCH | VERIFIED | RESEARCH | PUBLISHED MAR 29, 2026 ⚡ 0

Prognostic value of peri-operative circulating tumour DNA levels estimated by cell-free DNA methylation in patients with resectable colorectal liver metastases

This study evaluates the prognostic value of peri-operative circulating tumour DNA levels estimated by cell-free DNA methylation in patients with resectable colorectal liver metastases.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM | Stage IV | RESEARCH | VERIFIED | RESEARCH | PUBLISHED MAR 24, 2026 ⚡ 0

Appendiceal Goblet Cell Adenocarcinoma With Mismatch Repair Deficiency and Microsatellite Instability-High Status: A Novel Molecular Signature Guiding Immuno-Oncology Strategy

This study evaluates appendiceal goblet cell adenocarcinoma with mismatch repair deficiency and microsatellite instability-high status as a novel molecular signature guiding immuno-oncology strategy.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM | Early Detection | EARLY DETECTION | VERIFIED | EARLY DETECTION | PUBLISHED MAR 14, 2026 ⚡ 0

Screening, Prognostic, and Predictive Molecular Tools for Colorectal Cancer: Recent Advances in the Classical Background

This study evaluates screening, prognostic, and predictive molecular tools for colorectal cancer, focusing on recent advances in the classical background.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the most significant biomarker testing research and guidelines in this report window.

GENOMIC PROFILING
The Evolving Role for Repeat Molecular Testing in Metastatic Colorectal Cancer
This study evaluates the evolving role for repeat molecular testing in metastatic colorectal cancer.

GUIDELINE
Circulating Tumor DNA as a Biomarker for Recurrence After Curative Resection of Colorectal Cancer Liver Metastases: A Systematic Review and Meta-Analysis
This study evaluates circulating tumor DNA as a biomarker for recurrence after curative resection of colorectal cancer liver metastases through a systematic review and meta-analysis.

LIQUID BIOPSY
A new paradigm in postoperative colorectal cancer surveillance: integrating advanced imaging and multi-omics
This study evaluates a new paradigm in postoperative colorectal cancer surveillance by integrating advanced imaging and multi-omics.

RESEARCH
ROME trial wasn’t built in a day: the decade-long journey to randomized evidence in precision oncology
This study evaluates the decade-long journey to randomized evidence in precision oncology as part of the ROME trial.

EARLY DETECTION
From mono- to multi-cellular in vitro models: reconstructing colorectal cancer complexity for translational and personalized applications
This study evaluates the transition from mono- to multi-cellular in vitro models for reconstructing colorectal cancer complexity for translational and personalized applications.

What this means for you

If you have been diagnosed with metastatic colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision. Ask specifically about KRAS, NRAS, BRAF V600E, MSI/MMR status, HER2, NTRK, and TMB.

Generated from structured biomarker research data · 2026-04-12 10:43:09

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

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About This Report

This report is generated automatically from publicly available medical databases including PubMed, ASCO Journal of Clinical Oncology, Annals of Oncology, and NCI Cancer.gov. Entries are classified using deterministic rule-based logic and scored by clinical significance. AI-generated summaries are grounded exclusively in the source data provided.

Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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